

1. Data Overview

The dataset contains sales transaction records from different branches, cities, customer types, products, and product categories. It represents a realistic business situation where a company needs to understand its sales performance based on transaction data.

The original dataset contains 253 sales transaction records and 8 main columns:

Column	Description
sale_id	Unique ID for each sales transaction
branch	The branch where the sale was made
city	The city where the transaction occurred
customer_type	Type of customer, such as Member, Normal, or Unknown
product_name	Name of the product sold
product_category	Category of the product
quantity	Number of units sold
total_price	Total value of the sales transaction

The dataset includes both categorical and numerical variables.

Categorical variables such as 'city', 'customer_type', and 'product_category' are useful for comparing different business segments. Numerical variables such as 'quantity' and 'total_price' are used to measure sales volume and revenue performance.

I selected this dataset because it allows me to connect data analysis with real business decisions. For example, if a business knows which product categories and customer groups perform better, it can make better decisions about inventory, promotion, and customer loyalty programs.

	A	B	C	D	E	F	G	H	I
1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price	
2	1	A	New York	Member	Shampoo	Personal Care	3	17,66	
3	2	B	Los Angeles	Normal	Notebook	Stationery	10	29,43	
4	3	A	New York	Member	Apple	Fruits	15	19,26	
5	4	A	Chicago	Normal	Detergent	Household	5	41,73	
6	5	B	Los Angeles	Member	Orange Juice	Beverages	7	26,22	
7	6	A	Chicago	Normal	Shampoo	Stationery	9	108,24	
8	7	A	Chicago	Normal	Shampoo	Personal Care	1	11,46	
9	8	B	Los Angeles	Normal	Shampoo	Household	9	175,55	
10	9	A	Chicago	Member	Apple	Fruits	20	302,81	
11	10	B	Los Angeles	Member	Shampoo	Fruits	19	374,48	
12	11	A	Chicago	Normal	Detergent	Beverages	7	69,81	
13	13	A	New York	Normal	Orange Juice	Household	4	14,08	
14	12	B	Los Angeles	Member	Orange Juice	Household	12	88,47	
15	13	A	New York	Normal	Orange Juice	Household	4	14,08	
16	14	B	Los Angeles	Member	Apple	Fruits	5	47,13	
17	15	B	Los Angeles	Normal	Apple	Beverages	3	62,53	
18	16	B	Los Angeles	Normal	Notebook	Stationery	8	47,51	
19	17	B	Los Angeles	Member	Apple	Personal Care	2	4,56	
20	18	B	Los Angeles	Normal	Notebook	Fruits	15	212,82	
21	19	A	New York	Member	Apple	Beverages	2	41,92	
22	20	A	Chicago	Normal	Detergent	Personal Care	11	59,75	
23	21	B	Los Angeles	Normal	Shampoo	Personal Care	11	51,32	
24	22	B	Los Angeles	Normal	Orange Juice	Beverages	1	7,28	
25	23	A	New York	Member	Shampoo	Beverages	16	33,38	
26	24	A	Chicago	Normal	Apple	Fruits	2	40,77	
27	25	B	Los Angeles	Member	Detergent	Beverages	18	218,22	
28	26	B	Los Angeles	Normal	Notebook	Fruits	16	238,65	
29	26	B	Los Angeles	Normal	Notebook	Fruits	16	238,65	
30	27	A	New York	Member	Detergent	Household	2	20,44	
31	28	A	New York	Member	Detergent	Personal Care	17	299,41	
32	29	B	Los Angeles	Member	Shampoo	Stationery	17	223,74	
33	30	A	Chicago	Member	Orange Juice	Unknown	15	198,54	
34	31	A	Chicago	Unknown	Shampoo	Fruits	3	11,88	
35	32	A	New York	Member	Orange Juice	Unknown	3	17,3	
36	33	B	Los Angeles	Normal	Notebook	Stationery	17	126,24	
37	34	B	Los Angeles	Normal	Orange Juice	Household	9	78,29	
38	35	A	New York	Unknown	Apple	Personal Care	2	28,72	
39	36	A	New York	Member	Detergent	Personal Care	15	210,9	
40	37	B	Los Angeles	Member	Shampoo	Fruits	16	190,72	
41	38	B	Los Angeles	Member	Notebook	Beverages	14	81,19	
42	40	A	New York	Normal	Apple	Fruits	11	13,77	
43	39	A	New York	Member	Notebook	Beverages	15	246,21	
44	40	A	New York	Normal	Apple	Fruits	11	13,77	
45	41	A	New York	Member	Notebook	Household	10	115,56	
46	42	A	New York	Normal	Apple	Household	12	140,98	
47	43	A	New York	Normal	Notebook	Fruits	2	8,92	
48	44	A	Chicago	Unknown	Detergent	Fruits	11	43,08	
49	45	B	Los Angeles	Member	Shampoo	Household	8	134,65	
50	46	A	New York	Member	Shampoo	Fruits	8	136,53	
51	47	A	New York	Normal	Shampoo	Stationery	8	10,96	
52	48	B	Los Angeles	Normal	Detergent	Personal Care	20	213,57	
53	49	B	Los Angeles	Normal	Detergent	Unknown	2	13,5	
54	50	A	Chicago	Normal	Apple	Stationery	19	335,45	
55	51	A	Chicago	Normal	Detergent	Stationery	19	346,02	
56	52	A	Chicaao	Member	Orange Juice	Fruits	5	59,71	

2. Data Cleaning

Before doing the analysis, I cleaned the dataset carefully. This step is important because incorrect or duplicated data can lead to misleading results. If duplicated transactions are not removed, total revenue will be overstated and the business may think its sales performance is stronger than it actually is.

Missing values

I first checked missing values in the main columns. I used Excel filters and missing value checks to see whether any cells were blank. The result showed that there were no blank missing values in the eight main columns.

Missing Vaues	
sale_id	0
branch	0
city	0
customer_type	0
product_name	0
product_category	0
quantity	0
total_price	0

Although there were no blank cells, I found some records marked as “Unknown”. I treated these values carefully because “Unknown” is not a blank value, but it still shows incomplete classification in the dataset.

Customer Type

For customer_type, there were 3 records marked as Unknown. I did not delete these rows because the transactions still have valid values for city, product, quantity, and total price. If I deleted them, I would lose real sales information.

Instead, I kept “Unknown” as a separate category. This is a better decision because it allows the business to keep the revenue record while still recognizing that customer classification

was incomplete. In a real business, this could happen when the cashier forgets to select the customer type or when the customer information is not recorded properly.

	A	B	C	D	E	F	G	H
1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
34	31	A	Chicago		Shampoo	Fruits	3	11,88
38	35	A	New York		Apple	Personal Care	2	28,72
48	44	A	Chicago		Detergent	Fruits		43,08
255								
256								

1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
34	31	A	Chicago	Unknown	Shampoo	Fruits	3	11,88
38	35	A	New York	Unknown	Apple	Personal Care	2	28,72
48	44	A	Chicago	Unknown	Detergent	Fruits		43,08
255								

Product Category

For product_category, there were 6 records marked as Unknown. Similar to customer type, I decided not to remove these records because they still represent real sales transactions. However, I treated them separately in the analysis so that they would not be mixed with known categories such as Fruits, Beverages, Stationery, Household, and Personal Care.

This step is important because product category analysis is used for business decisions such as inventory planning and promotion strategy. If unknown categories are ignored, the total revenue may become inaccurate. If they are mixed into another category without evidence, the analysis may become misleading.

1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
33	30	A	Chicago	Member	Orange Juice		15	198,54
35	32	A	New York	Member	Orange Juice		3	17,3
53	49	B	Los Angeles	Normal	Detergent		2	13,5
72	68	A	New York	Normal	Detergent		1	7,22
91	87	A	Chicago	Member	Shampoo		2	5,86
103	99	A	New York	Normal	Detergent		17	277,4
255								
256								

1	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
33	30	A	Chicago	Member	Orange Juice	Unknown	15	198,54
35	32	A	New York	Member	Orange Juice	Unknown	3	17,3
53	49	B	Los Angeles	Normal	Detergent	Unknown	2	13,5
72	68	A	New York	Normal	Detergent	Unknown	1	7,22
91	87	A	Chicago	Member	Shampoo	Unknown	2	5,86
103	99	A	New York	Normal	Detergent	Unknown	17	277,4
255								

Quantity

For quantity, I checked whether there were blank or abnormal missing values. There were no missing values in the quantity column. I also calculated the median quantity using Excel:

=MEDIAN(G:G) = 11

The median quantity is useful because if the quantity had missing values, the median would be a reasonable replacement method. The median is less affected by extreme values than the mean. In this dataset, no replacement was needed, but checking the median helped me understand the central purchasing behavior of customers.

	B	C	D	E	F	G	H	I
1	branch	city	customer_type	product_name	product_category	quantity	total_price	
22	A	Chicago	Normal	Detergent	Personal Care		59.75	
48	A	Chicago	Unknown	Detergent	Fruits		43.08	
65	A	New York	Member	Detergent	Household		20.09	
255								

	B	C	D	E	F	G	H	I
1	branch	city	customer_type	product_name	product_category	quantity	total_price	
22	A	Chicago	Normal	Detergent	Personal Care	11	59.75	
48	A	Chicago	Unknown	Detergent	Fruits	11	43.08	
65	A	New York	Member	Detergent	Household	11	20.09	
255								

Duplicate Rows

After checking missing values, I checked for duplicate rows using Excel's Remove Duplicates function. I selected the full transaction table so that Excel would detect exact duplicate transaction records.

The dataset originally had 253 records. After checking duplicates, I found 3 exact duplicate rows.

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Alert

3 duplicate values found and removed;
250 unique values remain. Note that
counts may include empty cells, spaces,
etc.

OK

product_name	product_category	quantity	total_price	sale_id	branch	city	customer_type	product_name	product_category	quantity	total_price
oo	Personal Care	3	17,66								
ok	Stationery	10	29,43								
	Fruits	15	19,26								
nt	Household	5	41,73								
Juice	Beverages	7	26,22								
oo	Stationery	9	108,24								
oo	Personal Care	1	11,46								
oo	Household	9	175,55								
	Fruits	20	302,81								
oo	Fruits	19	374,48								
	Beverages	7	69,81								
	Household	4	14,08								
	Household	12	88,47								
	Fruits	5	47,13								
	Beverages	3	62,53								
	Stationery	8	47,51								
	Personal Care	2	4,56								
	Fruits	15	212,82								
	Beverages	2	41,92								
	Personal Care		59,75								
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	Household	2	20,44								
	Personal Care	17	299,41								
	Stationery	17	223,74								
	Unknown	15	198,54								
	Fruits	3	11,88								
	Unknown	3	17,3								
	Stationery	17	126,24								
	Household	9	78,29								
	Personal Care	2	28,72								
	Personal Care	15	210,9								
	Fruits	16	190,72								

3. Descriptive Statistics

After cleaning the data, I calculated descriptive statistics for quantity and total_price.

The average quantity per transaction is 10.62, while the median quantity is 11. This shows that a typical transaction contains around 10 to 11 units.

The average transaction value is 124.19, while the median is 95.43. This means some high-value transactions pull the average upward.

<i>quantity</i>		<i>total_price</i>	
Mean	10,61264822	Mean	123,7659289
Standard Error	0,375926392	Standard Error	6,481325292
Median	11	Median	94,17
Mode	2	Mode	14,08
Standard Deviation	5,97947532	Standard Deviation	103,0917898
Sample Variance	35,7541251	Sample Variance	10627,91712
Kurtosis	-1,257327406	Kurtosis	0,033590636
Skewness	-0,072949934	Skewness	0,907731819
Range	19	Range	424,96
Minimum	1	Minimum	2,18
Maximum	20	Maximum	427,14
Sum	2685	Sum	31312,78
Count	253	Count	253

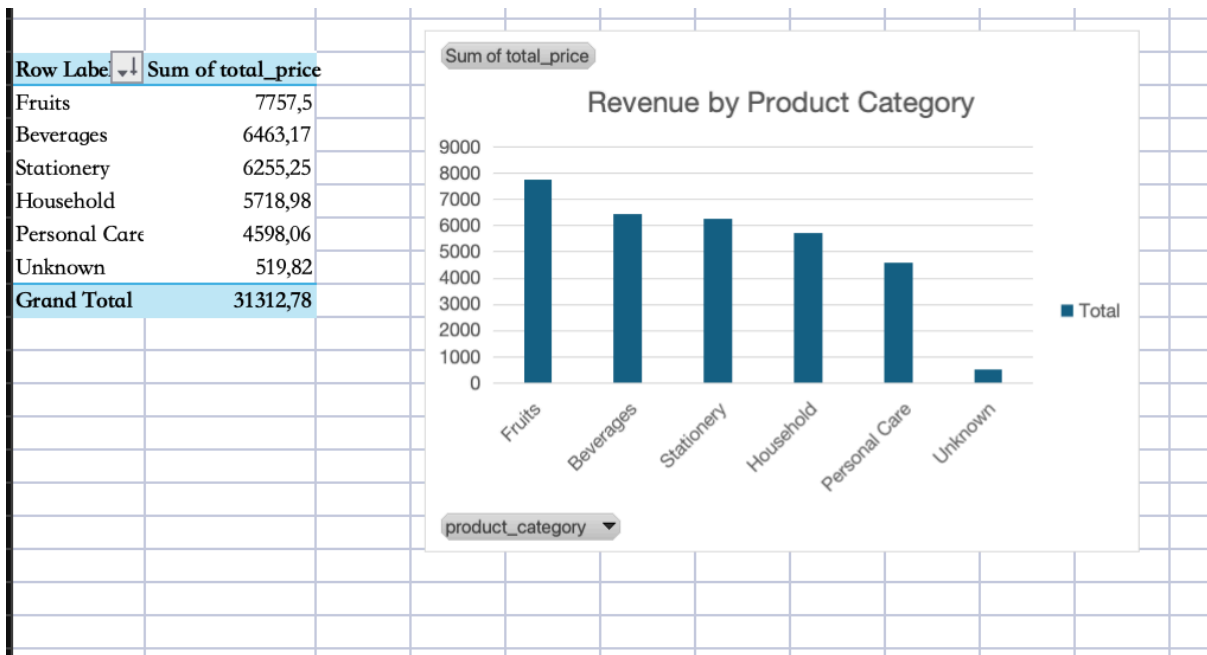
4. Insights

Insight 1: Revenue by Product Category

Fruits generated the highest total revenue, with 7,505.08, accounting for 24.17% of total revenue. This means Fruits is an important category for overall sales performance.

However, Stationery had the highest average revenue per transaction, at 142.16 . This shows that total revenue and average transaction value can tell different business stories.

A practical decision would be to maintain strong inventory for Fruits while using Stationery and Beverages for bundle promotions or cross-selling.



Insight 2: Revenue by Customer Type

Member customers generated 17,855.87, which accounts for 57.51% of total revenue. Normal customers generated 13,106.73, equal to 42.22% of total revenue.

Member customers also had a higher average transaction value, spending 136.30 per transaction compared with 112.99 for Normal customers.

This suggests that the membership group is valuable to the business. The company should maintain current members and encourage Normal customers to become members through reward points, discounts, or loyalty benefits.

Row Label ▼	Sum of total_price
Member	17855,87
Normal	13373,23
Unknown	83,68
Grand Total	31312,78

